

Three observer models — step-by-step comparison

A single-view comparison of the three observer models built for this project, aligned stage by stage. Each model's own equation labels are given in full in its individual reference:

Model	File	Equation labels
Switch — paper's Switching observer (Laquittaine & Gardner 2018)	switching_observer.py	H1–H5
HB - Rachel — our hb_integration observer (the abstract's model)	hb_integration.py	M1–M5
HB - Salma — independent hierarchical observer	hierarchical_confidence/model.py	B1–B5

The three do not number 1:1 (the Switch keeps evidence and prior as two separate read-outs; the two hierarchical models integrate them), so the tables below align on **conceptual stages** rather than on each model's internal step number.

Stage-by-stage comparison

Stage 1 — Prior representation

Aspect	Switch	HB - Rachel	HB - Salma
Form	Delta at 225° (prior read-out)	$\alpha \cdot V(\theta; 225, \kappa) + (1-\alpha)/360$	Pure $V(\theta; 225, \kappa)$, per κ
Mixture floor (uniform)	via lapse only	yes, weight $(1-\alpha)$	no ($\kappa=0$ grid point only)
Prior width set by	fitted k_{prior} per block	learned belief $b_t(\kappa)$	learned belief $H_t(\kappa)$
Label	H2	M1	B1

Stage 2 — Sensory likelihood

Aspect	Switch	HB - Rachel	HB - Salma
Form	$V(\cdot; \theta_{\text{true}}, k_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, k_{\text{like}}(c))$	$V(\cdot; \theta_{\text{true}}, k_{\text{sensory}}(c))$
Per-coherence reliability	3 k_{like}	3 k_{like}	3 k_{sensory}
Label	H1	M2	B2 (grid)

Stage 3 — How prior and evidence are combined (the defining axis)

Aspect	Switch	HB - Rachel	HB - Salma
Rule	MIXTURE — commit to prior OR evidence per trial	INTEGRATE, then average read-outs	INTEGRATE into one prior, then read out
Formula	$P_e \cdot \text{evidence} + P_{\text{prior}} \cdot \text{prior} + P_{\text{rand}} \cdot \text{unif}$	$\sum_{\kappa} b(\kappa) \cdot R_{\kappa}$ (avg of per- κ read-outs)	one posterior from $\sum_{\kappa} H(\kappa) \cdot V$, then MAP
κ marginalised...	n/a (no belief)	AFTER read-out	BEFORE read-out
Bimodality is...	imposed by stochastic commit	emergent from uniform floor	emergent from integrated posterior

Label	H3+H4	M4	B3+B4
-------	-------	----	-------

Stage 4 — Reliance-on-prior weight

Aspect	Switch	HB - Rachel	HB - Salma
What sets it	derived ratio $k_{\text{prior}}/(k_{\text{prior}}+k_{\text{e}})$ (Eq. 6)	fitted α (fixed across blocks)	emergent from belief (no explicit weight)
Free parameter?	no (derived)	yes (α)	no
Label	H3	M1	—

Stage 5 — Learned latent (trial-by-trial dynamics)

Aspect	Switch	HB - Rachel	HB - Salma
Learns across trials?	no — k_{prior} fixed per block	yes — belief $b_t(\kappa)$	yes — belief $H_t(\kappa)$
Update rule	none	forget then Bayes-correct on feedback	forget then Bayes-correct on feedback
Forgetting law	—	linear toward b_0 : $(1-\lambda)b + \lambda \cdot b_0$	geometric : $\log H \leftarrow \rho \cdot \log H$
Reset at block edge?	n/a	no (carried)	no (carried)
Label	—	M5	B2

Stage 6 — Read-out type

Aspect	Switch	HB - Rachel	HB - Salma
Read-out	MAP (Girshick push-forward)	MAP per κ (Girshick)	MAP, tie-aware
Tie handling	argmax	argmax	ties share mass 1/#ties
Label	H1/H2	M3	B4

Stage 7 — Motor noise + lapse

Aspect	Switch	HB - Rachel	HB - Salma
Motor	■ $V(\cdot;0,k_{\text{motor}})$	■ $V(\cdot;0,k_{\text{motor}})$	■ $V(\cdot;0,k_{\text{motor}})$ via FFT
Lapse	renormalised into Eq. 6 weights	$(\text{percept} + p_{\text{random}} \cdot \text{unif})/(1+p_{\text{random}})$	$(1-\text{lapse}) \cdot \text{percept} + \text{lapse}/n_{\theta}$
Label	H5	M4	B5

Numerical grid & fitting

Aspect	Switch	HB - Rachel	HB - Salma
Direction grid	360 (1°)	360 (1°)	72 (5°)
κ grid	n/a	~15 points	16 points (incl. $\kappa=0$)

Compute	linear space	linear space	log space (logsumexp)
Optimiser	(fitter-dependent)	(fitter-dependent)	Powell + Latin-hypercube multistart

Parameter counts

Model	N	Parameters
Switch	9	3 k_like + 4 k_prior + k_motor + p_random
HB - Rachel	7	3 k_like + α + k_motor + p_random + λ
HB - Salma	6	3 k_sensory + ρ + k_motor + lapse

In all three, the sensory reliabilities (k_like / k_sensory) and motor+lapse are shared in spirit. They differ in the prior machinery: the Switch spends 4 parameters on fixed per-block prior widths; HB - Rachel replaces those with 1 learned belief + α + λ ; HB - Salma replaces them with 1 learned belief + ρ and drops α .

One-line summary of the differences

- Switch vs the two hierarchical models:** the Switch *commits* to prior-or- evidence each trial (mixture, imposed bimodality) and its prior width is a *fixed* fitted constant per block. Both hierarchical models *integrate* and *learn* the prior width online.
- HB - Rachel vs HB - Salma (the two hierarchical models):** same idea, three definitional differences — (1) HB - Rachel averages per- κ read-outs *after* the fact while HB - Salma integrates the belief into *one* prior *before* reading out; (2) HB - Rachel keeps a fitted mixture weight α while HB - Salma's prior is a pure von Mises (no α); (3) HB - Rachel forgets linearly, HB - Salma geometrically. Empirically these agree near the prior and diverge in the far-from-prior (bimodality) regime — mean total-variation ≈ 0.16 in the far band, ≈ 0.06 near.